

Variable Selection and Model Prediction Based on Lasso, Adaptive Lasso and Elastic Net

Lei Fan

School of Information System and Management, National
University of Defense Technology
Changsha, China
fanleihjy@163.com

Shuai Chen

School of Information System and Management, National
University of Defense Technology
Changsha, China
15874141587@163.com

Qun Li

School of Information System and Management, National
University of Defense Technology
Changsha, China
liqun@nudt.edu.cn

Zhouli Zhu

School of Information System and Management, National
University of Defense Technology
Changsha, China
670663532@qq.com

Abstract—Statistics and modeling is gaining more and more attention as the time for big data is coming. Variable choosing plays a significant role during modeling. The traditional methods like OLS and ridge regression could not satisfy interpretability and prediction accuracy at the same time. Tibshirani. R prompted Lasso and the new method could not only solve the above problem but also decrease the complexity of calculation. The paper aims to compare Lasso and adaptive lasso, elastic net. We do experiment on the classical case—diabetes patient data and choose model with AIC, BIC and cross validation to get the advantage and disadvantage of the above methods. According to the result, we draw the conclusion that Lasso performances better in variable selection, however, the predictions are not as accurate as the other two methods. Adaptive Lasso selects less variables and gives a more accurate prediction value. Elastic net has a most interpretative model.

Keywords—variable selection; prediction accuracy; Lasso; adaptive lasso; elastic net.

I. INTRODUCTION

With the coming of big data era, data mining is showing its importance and has had a place in the math world. In the field, model selection is crucial and dimension reduction is the first step to choose a proper model. At the beginning of modeling, we always select as many variables as possible to reduce the deviation. However, what we need is a most interpretative model in practice. A selected model should satisfy the follow criterions: (1)prediction accuracy, (2)model interpretability, (3)model stability, (4)calculation complexity[1].

Traditional OLS gives a model with interpretability, however, the model is always over-fitting with a poor performance in prediction accuracy. Ridge regression gives a shrinkage of regression coefficients. But the model is a continuous process and could not reduce the dimension of variables, which results in a bad interpretative model. In 1996, Tibshirani. R prompted a new method named Lasso based on the two methods above, which solves the problems and satisfies the model selecting criteria[2]. Likely, Lasso has its

limit. Large amount of scholars have done research on the method and prompted series of improvements. In the paper, I choose Lasso, Adaptive Lasso and Elastic net to do experiment on the classical case—diabetes patient data and analyze results to distinguish their performance on variable selection and prediction accuracy.

II. METHOD INTRODUCTION

A. Ridge Regression

Ridge regression[3] was a biased estimate prompted by Hotel and Kennard in 1970, coming from OLS(ordinary least squares). We define

$$\hat{\beta}_{ridge} = \arg \min \left\{ \|Y - X\beta\|^2 + \lambda_1 \sum_j \beta_j^2 \right\} \quad (1)$$

Ridge regression is more stable than OLS. Ridge regression introduces L2-norm loss function and shrinks regression coefficients equally towards zero. It thus selects all variables with a poor interpretative model[4].

B. Lasso

Lasso was first prompted by Tibshirani.R. It obtains sparse linear model in high-dimensional variable space. Lasso estimator uses L1-penalty to shrink regression coefficients towards zero, thus the variable is selected. Lasso has been widely used in image identification, genetic selection and many other domains.

A linear regression model is defined as[5]

$$y_i = \alpha_i + \sum_{j=1}^p \beta_j x_{ij} \quad (2)$$

where x_{ij} are y_i prediction variable and its corresponding response variable. We assume each observation is independent and prediction variables are standardized. Then Lasso estimate is described as

$$\hat{\beta}_{lasso} = \arg \min \left\{ \|Y - X\beta\|^2 + \lambda \sum_j |\beta_j| \right\} \quad (3)$$

where β_j is a nonnegative regularization parameter and the Lasso shrinks the coefficients towards 0 as λ increases, and a model is chosen with a suitable λ [1].

The algorithm of Lasso is listed as follows:

Input: Dataset, including prediction and response variables

Output: Regression coefficients to be estimated in the model

Step 1: Data pretreatment, ensure data to satisfy standardization

Step 2: Adopt Lasso to train model and LARS[6] to calculate the solution path:

- a) calculate absolute value of regression coefficients and update the solution set.
- b) confirm the least angle direction
- c) calculate step length
- d) predict response variable

C. Adaptive Lasso

Lasso has advantage over ridge regression and subset selection, however, it shrinks the regression coefficients in the same degree and lacks the Oracle property. We assume that the quantity of variables increases with the magnification of sample capacity in a high-dimensional data model. The Oracle property[7] is

- (1) Consistency in variable selection;
- (2) Asymptotic normality.

Adaptive lasso[8] estimate is based the Oracle property and could be considered as the weighted Lasso:

$$\hat{\beta}_{alasso} = \arg \min \left\{ \|Y - X\beta\|^2 + \lambda \sum_j \omega_j |\beta_j| \right\} \quad (4)$$

where ω_j are the known weights for regression coefficients. In the paper, we use coefficients of OLS to define the weight value.

The algorithm of Adaptive lasso is listed as follows[1]:

Input: Dataset, including prediction and response variables

Output: Regression coefficients to be estimated in the model

Step 2: Data pretreatment, ensure data to satisfy standardization

Step 2: Set $x_j^* = x_j / \omega_j, j = 1, 2, \dots, p$

Step 3: For all λ , solve Lasso problem.

$$\hat{\beta}_{lasso} = \arg \min \left\{ \|Y - X\beta\|^2 + \lambda \sum_j |\beta_j| \right\}$$

Step 4: Give the output:

$$\hat{\beta}^{**} = \arg \min \left\{ \|Y - X\beta\|^2 + \lambda \sum_j \hat{\omega}_j |\beta_j| \right\}$$

D. Elastic Net

Lasso is not robust to extreme correlations among the predictors. It has three shortcomings. p is the number of prediction variable and n is the number of response variable[9].

(1) When $p \gg n$, Lasso could only select variable less than n , which leads to an excessive sparse model.

(2) If the prediction variables could be divided into several groups and variable in a group has an intense interaction, Lasso selects only one variable in each group.

(3) When there are collinear variables, Lasso estimates are similar to ridge regression, as it selects the model which has a poor performance in prediction accuracy.

Zou and Hastie prompted an expansion method named Elastic net to select variable. Elastic net[10] combines Lasso and ridge regression, that is, it uses a mixture of L1-penalty and L2-penalty. Elastic net was proposed:

$$\bar{\beta}_{elasticnet} = \arg \min \left\{ \|Y - X\beta\|^2 + \lambda_1 \sum_j |\beta_j| + \lambda_2 \sum_j \beta_j^2 \right\} \quad (5)$$

where $\lambda_1 \sum_j |\beta_j| + \lambda_2 \sum_j \beta_j^2 \leq t$.

III. MODEL SELECTION CRITERION

A. AIC (Akaike Information Criterion)

AIC[11] was founded and developed by Japanese statistician H. Akaike. AIC is based on the concept of entropy, and could balance the complexity of model and superiority of data match. AIC is defined as: $AIC = 2k - 2 \ln(L)$, where k is the quantity of parameter and L is the likelihood function. The superiority of match is developed while the number of free parameter increases.

B. BIC (Bayes Information Criterion)

BIC[12] is used when information is incomplete. Firstly, unknown parameter is estimated with subjective probability and priori probability is obtained. Secondly, update the subjective probability with sample data and Bayes formula to

gain posterior probability. Finally, use expectation and posterior probability to make an optimal decision. BIC is defined as: $BIC = 2\ln(k) - 2\ln(L)$, where k is the quantity of parameter and L is the likelihood function.

C. Cross Validation

Cross validation[13] is to divide data into two parts: one part is to train model, we always name it training set. The other one is test set to test the trained model. In practical application, limited quantity of available data pushes us to reuse the data. We divide the data several times to obtain a proper estimation. There are many different forms of cross validation and we introduce V-fold cross validation[14] here. V-fold cross validation divides the dataset into V parts and takes one part as test set each time, while the remained V-1 parts are training set. Repeat V times and the estimation of generalization error is the mean of the V times.

IV. CASE ANALYZE AND RESULTS

A. Data Specification

We choose the classic case-diabetes dataset[15]to analyze Lasso and its related methods. We compare the analysis results

of variable selection and model prediction. The data comes from the website of Stanford University: www.stat.stanford.edu/~hastie/Papers/. The methods are using the lars[6], ridge[16] and msgps[15] package in R provided in the Internet and Matlab program.

In the diabetes dataset, there are 10 basic variables including age, gender, BMI, blood pressure and six serum measured values. The response is a quantitative measure of disease progression one year after baseline[15].

B. Modeling

The experiment aims to do the modeling with training set and predicts response variable of test set. The diabetes dataset has 442 groups of data and the previous 400 as training set while the remaining 42 as test set. Firstly, standardize the prediction variables according to the algorithm, which eliminates dimension of different data types. Then we use R to analyze the data and thus obtain the best tuning parameter from Matlab and prediction values of response. LARS algorithm is used to compute Lasso, Adaptive Lasso and Elastic net. For each competitor, tuning parameter is selected by model selection criterion: AIC, BIC and V-fold CV.

TABLE I. THE ESTIMATED COEFFICIENTS FOR THE DIABETES DATA BASED ON THE LASSO, ADAPTIVE LASSO($\Gamma=1$) AND ELASTIC NET ($\alpha = 0.5$) WITH AIC AND BIC.

Variable number	Ridge		Lasso		Adaptive Lasso		Elastic net	
	AIC	BIC	AIC	BIC	AIC	BIC	AIC	BIC
9	80.1346	50.9876	45.5804	42.8016	78.3087	46.5839	76.2246	40.2157
2	-24.0270	-20.6606	-18.2417	-14.9893	-22.3628	-20.4437	-21.7163	-18.9084
3	6.0012	5.8820	5.9494	5.8488	5.8648	6.1094	5.9243	5.5974
4	1.2519	1.1274	1.0547	0.9847	1.1443	0.9752	1.1480	1.0875
7	-0.7522	-0.5610	-0.8276	-0.7823	0.5912	-1.1029	0.4953	-0.7043
5	-0.4920	-0.4157	-0.1525	-0.0915	-1.4587	-0.0743	-1.3736	-0.1364
8	1.9130	1.8612	0.0000	0.0000	4.1941	-2.6711	3.7255	4.2019
6	0.3154	0.2070	0.0000	0.0000	1.1573	0.0000	1.0873	-0.0836
10	0.0310	0.0240	0.0123	0.0000	0.0000	0.0000	0.0261	0.0221
1	-0.0701	-0.0606	0.0000	0.0000	0.0000	0.0000	-0.0869	-0.0246

C. Variable Selection

Different regression coefficients are supposed to be obtained when different model selection criterion is chosen. However, AIC and V-fold CV have the same result. Thus, results given by V-fold CV are no longer considered and other results are shown in Table 2. BIC performs better and the coefficients are smaller. Therefore, we choose BIC to select model in the following parts.

We have four modeling estimators: Ridge regression, Lasso, Adaptive Lasso and Elastic net. Ridge regression does not have

any zero coefficients. Lasso selects the least variables while Adaptive Lasso adds a variable comparedwith Lasso. Elastic net has three coefficients close to zero. We’ve known that during modeling, the more variables we save the higher degree of model matching. And it’s obvious when comparing degree of fitting: Ridge regression > Elastic net > Adaptive Lasso > Lasso. However, in model selection, high degree of fitting does not mean the best model, for if a model fits the training data excessively it will influence its predictive results[17].

Figure.1 shows how the degrees of freedom and coefficients vary with the sum of regression coefficients for

Lasso estimation. On the Lasso penalty, the degrees of freedom is six with the best tuning parameter about 1750. And it is obvious that the degree of freedom increases while the tuning parameter increases, however, when the value of β is larger than 1750 the speed of freedom degree increase has slowed. The left panel in Figure 1 presents coefficient values and they are matching what Table 2 shows. Figure.2 and Figure 3 give the results of Adaptive Lasso and Elastic net respectively. For

Adaptive Lasso, the best β value is about 600 and the corresponding degree of freedom is seven. The left panel in Figure 2 shows that when $\beta < 600$, the degree of freedom increases rapidly. And when $\beta > 600$, the growth rate declines. For the value of β is doubled, however, only one variable coefficient is shrunk to zero. Figure.3 shows result of Elastic net and we select seven variables for the method.

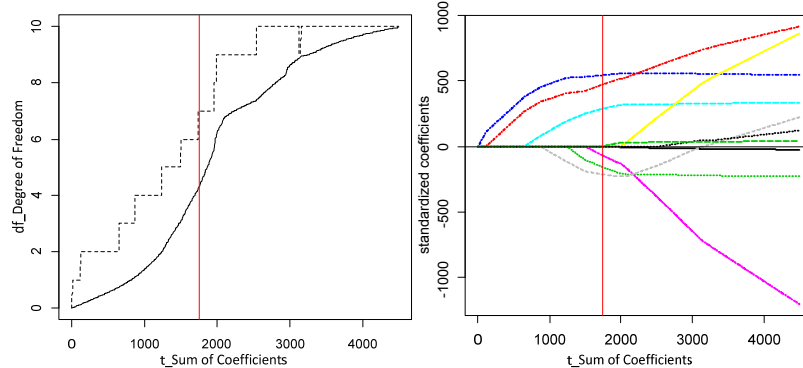


Fig.1. Lasso method: The degree of freedom(left panel) and solution path(right panel) and the red line in the right panel is the selected model by BIC.

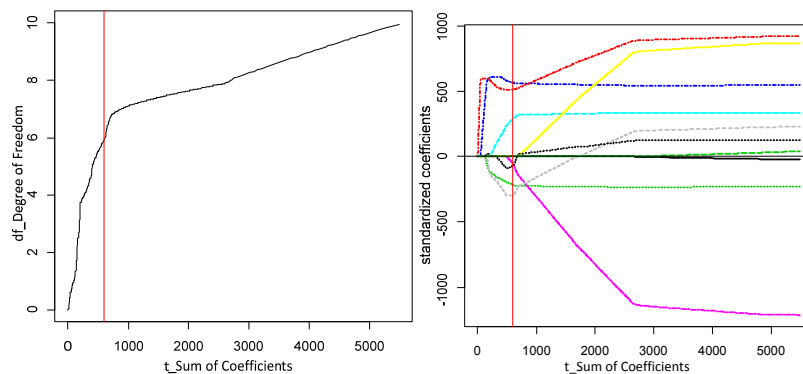


Fig.2. Adaptive Lasso method: The degree of freedom(left panel) and solution path(right panel) and the red line in the right panel is the selected model by BIC.

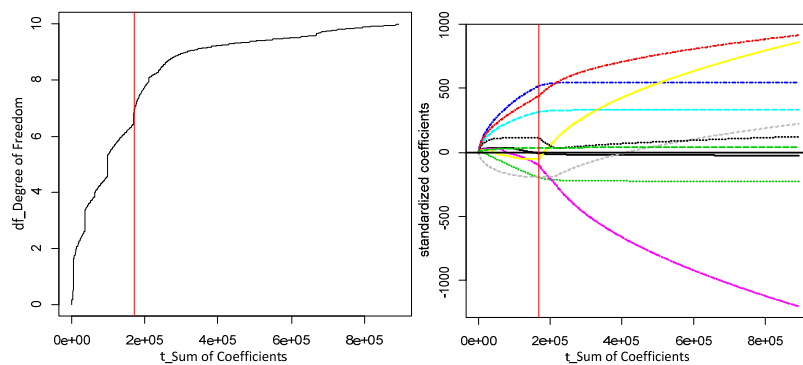


Fig.3. Elastic net method: The degree of freedom(left panel) and solution path(right panel) and the red line in the right panel is the selected model by BIC

D. Model Prediction

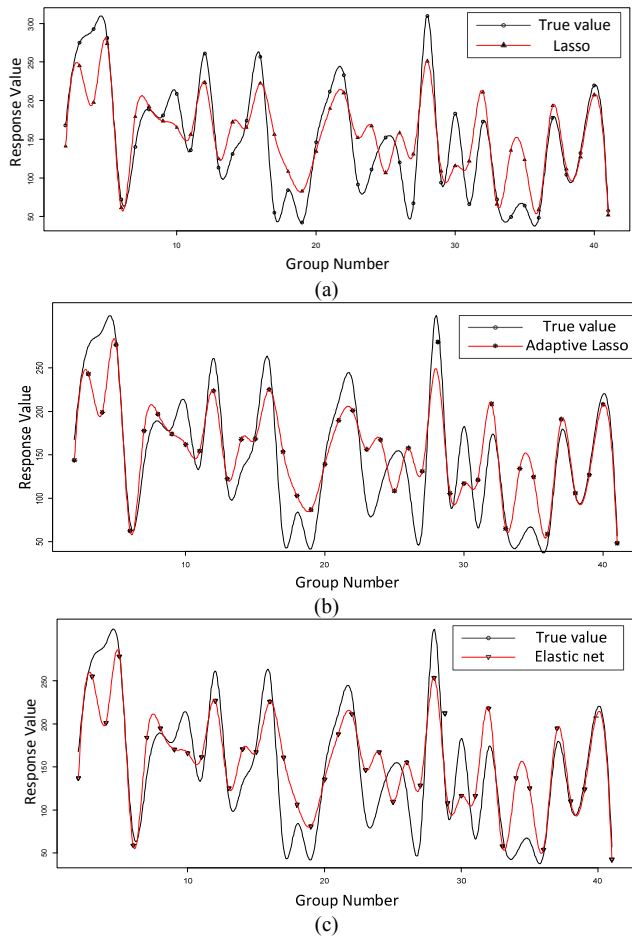


Fig.4. Prediction results of test set: (a)-(c) shows comparison of true value and prediction value of Lasso, Adaptive Lasso, Elastic net respectively

In this part we choose 400 groups data to modeling and predicts the response of remaining 42 groups data. We use the above three different methods to modeling and compares the results with the given response value. Figure 4(a)-(c) shows the comparison between the true value and the predict value by Lasso, Adaptive Lasso and Elastic net. Lasso estimation gives a less match model and prediction result is poorer than Elastic net estimation. According to the prediction value, we calculate the variance and it's 5.9899 for Lasso. Adaptive Lasso uses weight value to shrink coefficients in different degrees. Thus the method selects more variables than Lasso and gets a model with better data matching and poorer prediction accuracy. And the variance for Adaptive Lasso is 5.9904. Elastic net is a method that combines Lasso and ridge regression, which provides a better interpretative model. Therefore, the method has developed prediction accuracy with the variance 5.9759.

V. CASE ANALYZE AND RESULTS

In the paper, we compare the modeling and prediction function of Lasso, Adaptive Lasso and Elastic net. According to the results, Lasso obtains model with best data matching and Elastic net has most accurate response values. In the future,

more research should be done to make a sense whether the methods perform likely on large sample due to a small sample dataset is used in the experiment.

REFERENCES

- [1] Jianchao Gong. Application of Lasso and its related methods on the basis of generalized linear model [D]. Zhongnan University, 2008. DOI:doi:10.7666/d.y1327770.
- [2] Tibshirani, Robert. Regression shrinkage and selection via the lasso: a retrospective[J]. Journal of the Royal Statistical Society, 2011, 73(3):273–282.
- [3] Hoed A.E. and Kennard R.W., Ridge regression: biased estimation for non-orthogonal problems, *Techometrics*, 1970, 12:55-68.
- [4] Ogutu J O, Schulz-Streeck T, Piepho H P. Genomic selection using regularized linear regression models: ridge regression, lasso, elastic net and their extensions[J]. *Bmc Proceedings*, 2012, 6:S10(Suppl 2): S10.
- [5] P, Burton, L, Gurrin, P, Sly. Extending the simple linear regression model to account for correlated responses: An introduction to generalized estimating equations and multi-level mixed modelling[J]. *Statistics in Medicine*, 1998, 17(11):1261-1291.
- [6] Efron B, Tibshirani R. Least Angle Regression[J]. *Ann Statist*, 2004, 32(2):2004.
- [7] Fan J, Li R. Variable Selection via Nonconcave Penalized Likelihood and its Oracle Properties[J]. *Journal of the American Statistical Association*, 2001, 96(456):1348-1360.
- [8] Hui Zou. (2006), The Adaptive Lasso and Its Oracle Properties.
- [9] Yin Lu. Variable Selection Methods Via the Elastic Net in Generalized Linear Models [D]. Beijing Jiaotong University, 2011.
- [10] Hui Zou, Trevor Hastie. (2005), Regularization and variable selection via the elastic net.
- [11] Akaike H. Factor analysis and the AIC[J]. *Psychometrika*, 1987, 52:317-332.
- [12] Burnham K P. Understanding AIC and BIC in model selection[J]. *Sociological Methods & Research*, 2004, 33.
- [13] Kohavi R. A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection[J]. *ICML*, 1995, 14:1137--1143.
- [14] Burman P. A comparative study of ordinary cross-validation, v-fold cross-validation and the repeated learning-testing methods[J]. *Biometrika*, 1989, (3):503-514.
- [15] Hirose, K., Tateishi, S. and Konishi, S.. (2011). Efficient algorithm to select tuning parameters in sparse regression modeling with regularization. *arXiv:1109.2411* (arXiv).
- [16] Cule E, Vineis P, Iorio M D. Significance testing in ridge regression for genetic data[J]. *Bmc Bioinformatics*, 2011, 12(54):1-15.
- [17] Cawley G C, Talbot N L C. On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation[J]. *Journal of Machine Learning Research*, 2010, 11(1):2079-2107.